

Data visualization guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 0.2 (pilot) September 2026 All example data is synthetic

OVERVIEW

How to use this guide

One set of rules, three ways in. The guide teaches the thinking; the starter kit makes the right answer the easy one in every tool we use.

Every chart we make should do one job: answer a business question so clearly that the reader knows what to do next. The rules here are written to be checked, by a reviewer or by an agent, before a chart leaves the team.

Inspired by the structure of a leading consulting firm's public data-visualization guide, rebuilt for McKesson's brand and for the work of Workforce Innovation.

CONSULTANTS AND ANALYSTS

Decks, reports, workbooks

Read Fundamentals once, then use the chart pages as you build. Keep the cheat sheet next to you.

- PowerPoint template and starter deck
- Excel chart workbook with the palette
- One-page cheat sheet and chart chooser
- Power BI report theme

DEVELOPERS AND DATA SCIENTISTS

Python, HTML, R, Power BI

Each chart page links to tested code. Themes are generated from one token file, so every tool matches.

- `mckviz` for matplotlib and Plotly
- CSS variables, Plotly and Vega-Lite themes
- `theme_mck()` for ggplot2
- `mv.lint(fig)` to check a chart

CODING AGENTS

Claude, Copilot and others

Agents read `AGENTS.md` automatically in this repo. For other repos, copy one self-contained rules file.

- `AGENTS.md`: workflow and rules
- `mck-chart` skill: question to QA
- `dist/agents/mck-viz-rules.md`
- Cookbook pages with runnable snippets

The five steps behind every chart

1 Question. Write the question the chart answers in one sentence.

2 Chart. Pick the chart from the question type, not from habit.

3 Build. Grey first, then color only what the headline names.

4 Headline. Compute the finding, then write it as a sentence.

5 QA. Run the checklist and look at the rendered chart.

OVERVIEW

Contents

Fundamentals apply to every chart. Each chart family then gets the same treatment: anatomy, variants, best practice and how to build it.

This is the pilot release. Sections marked NEXT follow the same pattern once the look and depth of the pilot are agreed.

01 **Fundamentals**
Question first, how readers read, anatomy, headlines, color, type, clutter, honest scales, people data, accessibility

02 **Magnitude**
Bar and column charts: anatomy, variants, best practice, build it

03 Part-to-whole NEXT
100% stacked bar, waterfall, treemap, donut

04 Change over time NEXT
Line, stacked area, bubble timeline, small multiples

05 Ranking NEXT
Slope, bump, dot plot and dumbbell, heat table

06 Distribution NEXT
Histogram, strip, box plot

07 Correlation NEXT
Scatter and bubble, heatmaps, why not dual axes

08 Relationships and flows NEXT
Sankey, parallel coordinates, Venn

09 Geo-spatial NEXT
Choropleth, bubble map, dot density

10 Single number NEXT
KPI and big number, progress to target

11 Formats NEXT
Slides, reports and print, web and mobile, dashboards, interactivity, animation, social

12 Toolkit NEXT
Downloads, set-up for each tool, the QA checklist, guidance for agents

Start with the question, not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

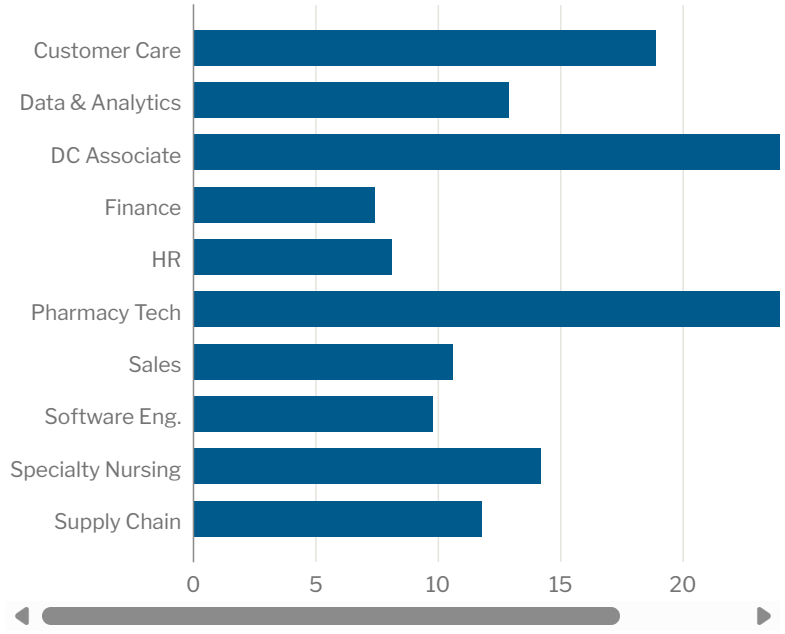
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

Attrition by job family

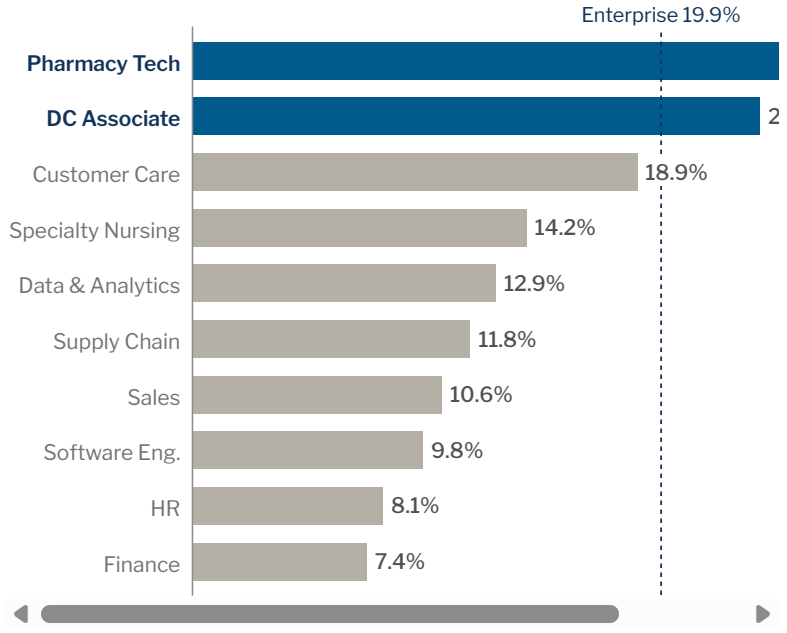
FY26



Shown
A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Explained
The headline states the finding, the order ranks the families, and color points to the two the headline names.

How readers read a chart

Readers move through a chart in three steps. Design for each one, in order.

1 Perceive

In the first second the eye takes in shape, length and color before reading a word. Make the thing that matters the thing that stands out: one color against grey.

2 Interpret

Next the reader looks for meaning: headline, labels, units, benchmarks. The headline should confirm what they just saw, or sharpen it.

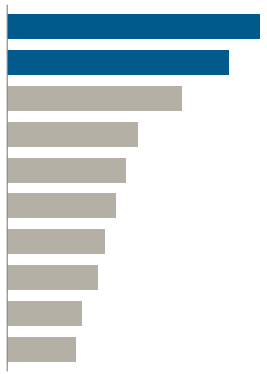
3 Act

Finally they ask "so what?". A callout that sizes the implication turns a finding into a decision.

STEP 1 · PERCEIVE

Two bars stand out

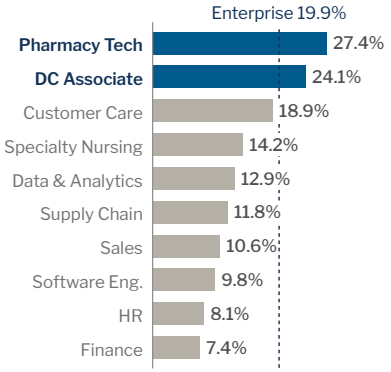
No words yet. Grey context and blue focus already tell the reader where to look.



STEP 2 · INTERPRET

Labels give it meaning

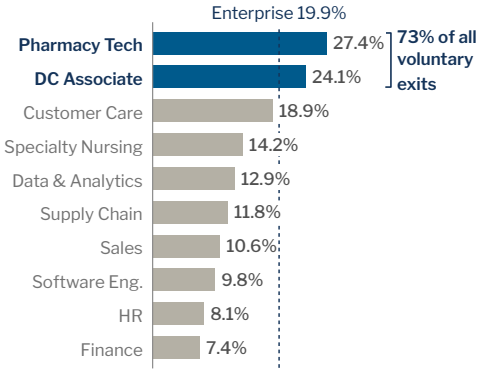
Names, values and a labelled benchmark show the two families sit well above 19.9%.



STEP 3 · ACT

The callout sizes the prize

Together they hold 58% of headcount but 73% of voluntary exits: a retention plan for two families moves the enterprise number.



Anatomy of a chart

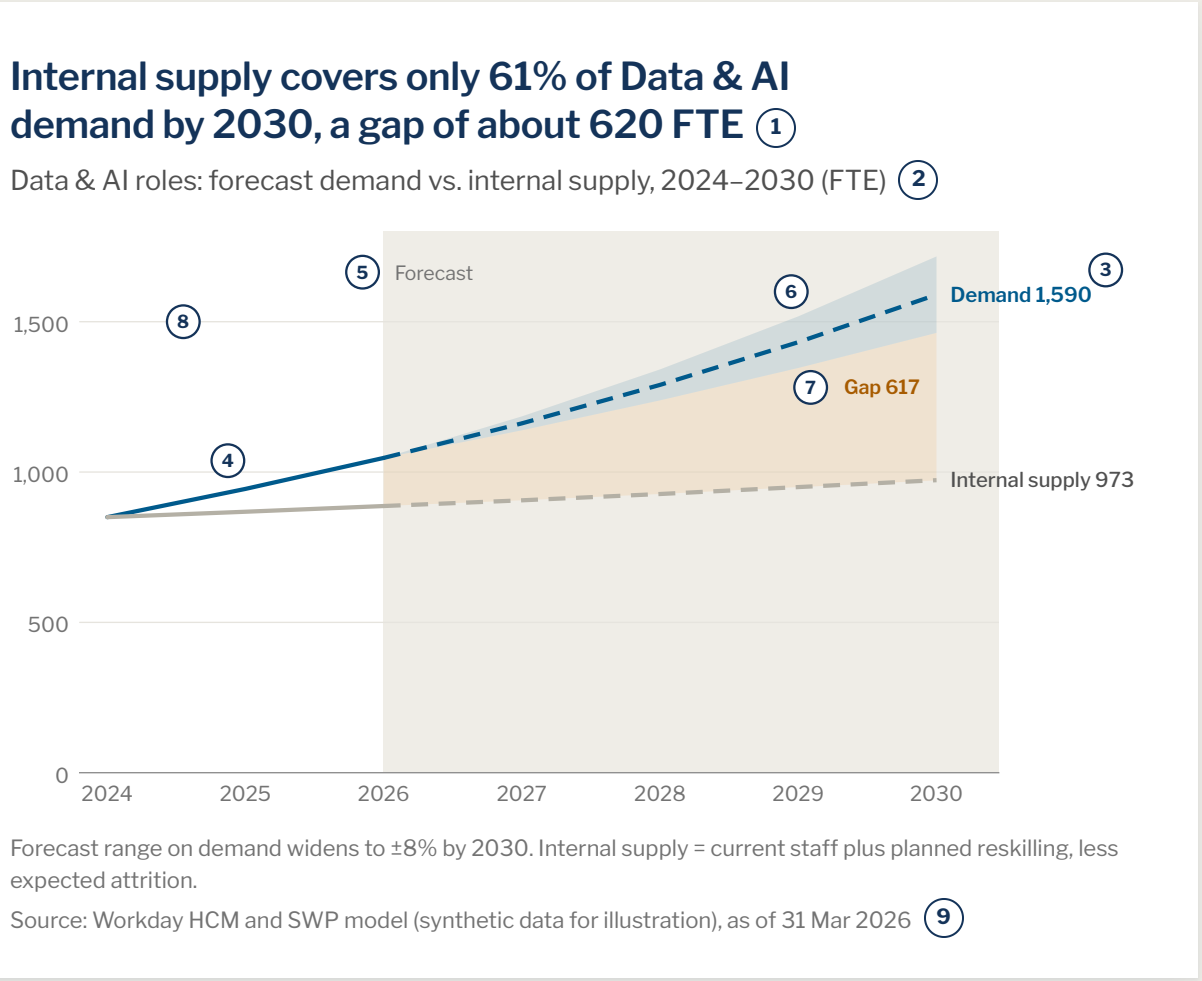
Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.



FUNDAMENTALS

Write the headline last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

The formula

Who or what + verb + number + so what.

Before you keep it

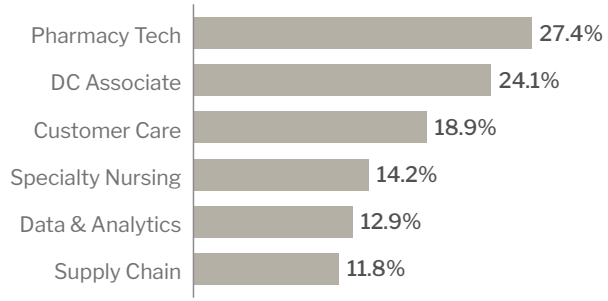
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

LABEL	Attrition by job family · names the topic; says nothing
VAGUE	Attrition varies across job families · true of every chart ever made
OPINION	Attrition is out of control in the frontline · a claim the chart can't support
INSIGHT	**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**
INSIGHT + ACTION	**Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate**

Attrition varies across job families

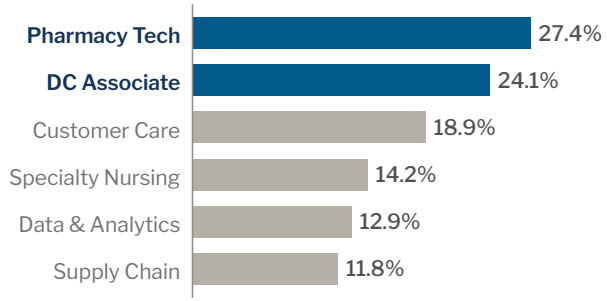
Voluntary attrition by job family, FY26 (%)



Headline and chart both neutral
Nothing tells the reader where to look.

Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)



Headline and highlight agree
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

Grey first, then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

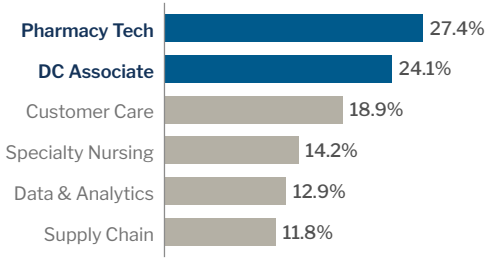
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

Orange means "worse". Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

Focus

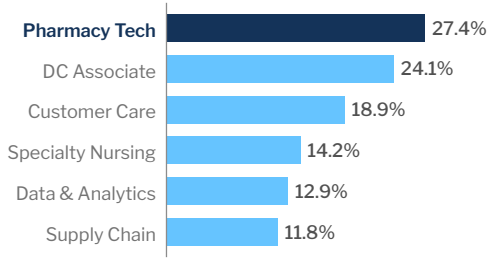
All grey; the items the headline names in blue.



PATTERN 2

Uniform + accent

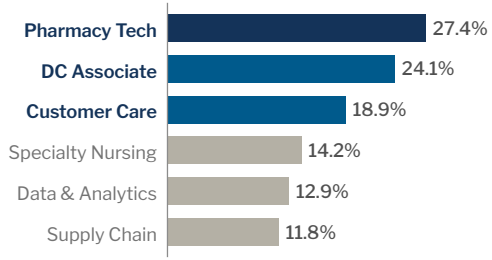
One series in light blue; the single exception in navy.



PATTERN 3

Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.



Color roles

Context · everything the reader should see but not focus on	`#B4B0A5`
Focus · what the headline names	`#005A8C`
Accent · the one exception that must pop (navy, not orange)	`#143359`
Light blue · the series when navy marks the exception	`#66C4FF`
Orange · only when it means worse: below target, shrinking, a shortfall	`#EF8200`
Total and benchmark · totals, targets, reference lines	`#143359`
Context light · de-emphasized fills in stacks and areas	`#DEDBD1`

Three palettes, three jobs

Pick the palette from what the color encodes.

Categories

Only when categories must be told apart. At most six, always in the same order, so "Full-time" is the same blue on every slide.

One measure, light to dark

For heat tables and choropleths of a single measure. Darker means more.

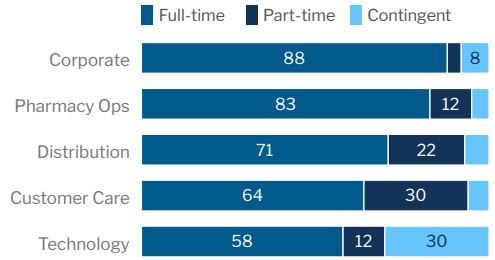
Gap to a benchmark

For values centered on zero or a target. Orange below, blue above, a neutral middle.

Text on a fill switches between white and navy, whichever contrasts more (`mv.text_on`).

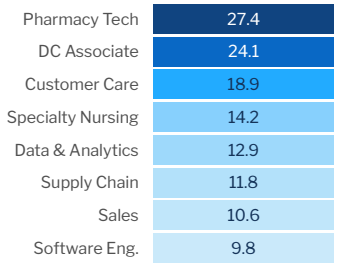
CATEGORICAL

Workforce mix by function (%)



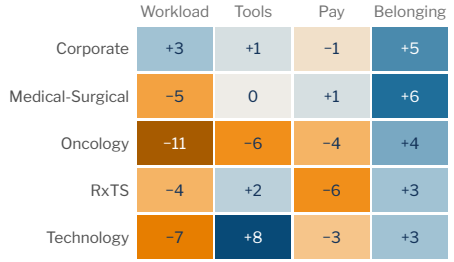
SEQUENTIAL

Voluntary attrition, FY26 (%)



DIVERGING

Engagement vs. healthcare benchmark (pts)



Categorical
Max 6, always this order

005A8C 143359 66C4FF 00AF9D 00737A EF8200

Sequential
One measure, light to dark

E6F3FB B6E2FA 66C4FF 009EFF 0A5FBC 143359

Diverging
Centered on 0 or a target

A75B00 EF8200 F7C180 EFEDE7 99BDD1 005A8C 143359

Type and proportion

One typeface, Libre Franklin, in a few fixed roles. Type should frame the data, not compete with it.

The headline is the largest text on the chart but the data is the largest *thing*. If the headline is the first thing you notice from across the room, it is too heavy.

- Nothing below 9 pt on slides, 11 px on screens
- Sentence case everywhere; no all-caps headlines
- Headline, subtitle, key, footnote and source share one left edge
- Big numbers use the light weight

Libre Franklin is open source (SIL OFL). Set-up for Office, the web and Power BI is in the Toolkit.

Role	Sample	Weight	Slide	Screen
Headline	**Attrition is concentrated in two job families**	Bold	24 pt	22 px
Subtitle	Voluntary attrition, FY26 (%)	Regular	14 pt	15 px
Data labels	27.4%	Medium	12 pt	13 px
Axis and category labels	Pharmacy Technician	Regular	11 pt	12 px
Callouts	Hiring freeze, Q2	Regular	12 pt	13 px
Footnote and source	Source: Workday HCM, as of 31 Mar 2026	Light	9 pt	11 px
Big number	75%	Light	54 pt	56 px

Proportion on a 16:9 slide

Margins of about 0.5 in. The chart takes at least 60% of the slide height. One message per chart, at most two charts per slide; the slide title is the headline.

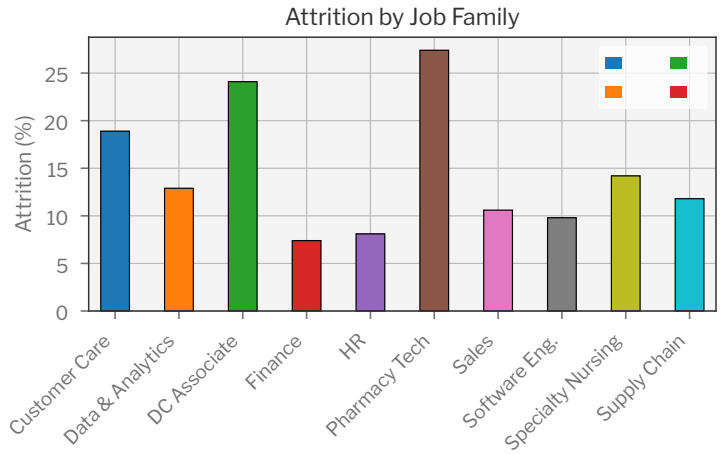
Proportion on a screen

The chart card sits on a quiet background with generous padding. Headline about 1.4× the subtitle. On phones, swap end labels for a top key and let wide exhibits scroll sideways rather than shrink.

Label directly, delete the rest

Every element must earn its place. If removing it loses nothing, remove it.

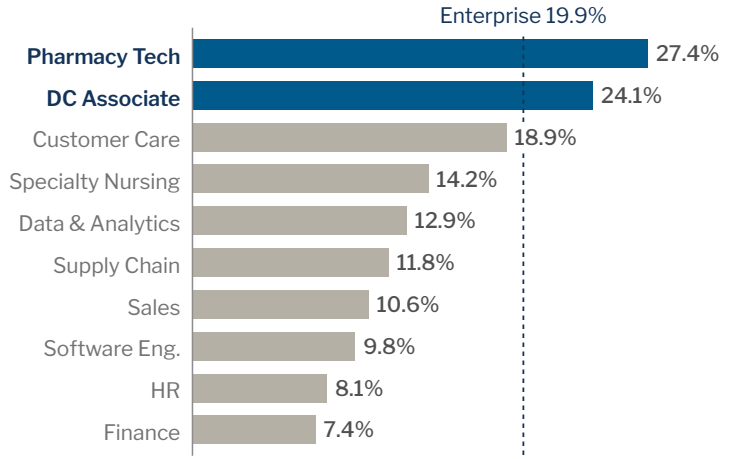
1. Replace the label-style title with an insight headline
2. Sort by value, not alphabet
3. Turn long-named columns into horizontal bars
4. Grey everything, then color the point
5. Label values in place; delete the value axis, gridlines and tick marks
6. Delete the legend, borders and background fill
7. Add the benchmark, labelled where it sits



Tool defaults
Rainbow fills, borders, rotated labels, gridlines, a legend that decodes nothing.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Decluttered
The same data. Fewer elements, and every one of them carries information.

FUNDAMENTALS

Honest scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

Bars start at zero. If the differences only show on a narrow range, change the chart, not the axis.

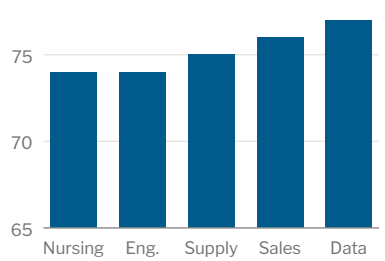
No dual axes. Two scales let you make any two lines cross wherever you like.

Bubble area, not diameter, is proportional to value.

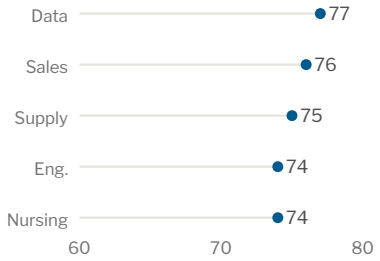
Pies take five slices at most, and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

Truncated bars

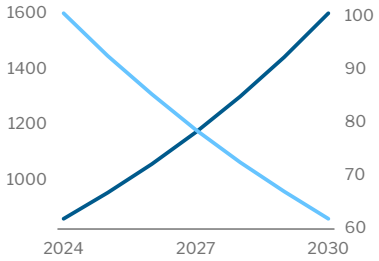


Try to avoid
An axis from 65 turns a 4% difference into a 33% one.

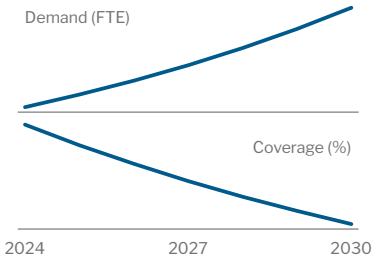


Recommended
A dot plot compares position, so the range can be narrow.

Dual axes

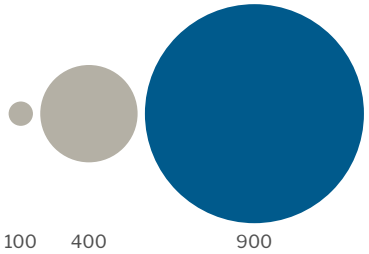


Try to avoid
The crossing point is an artifact of two chosen scales.

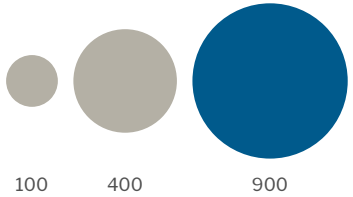


Recommended
Two aligned charts share the time axis.

Bubble size

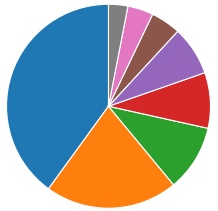


Try to avoid
Sized by diameter, 900 looks 81× bigger than 100.

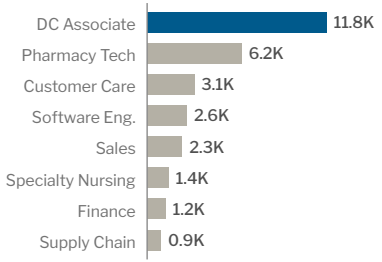


Recommended
Sized by area, 900 is 9× bigger, as it should be.

Too many slices



Try to avoid
Eight similar angles can't be ranked by eye.



Recommended
A sorted bar ranks them instantly.

People data is personal

Workforce charts describe colleagues, and carry extra obligations.

Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

Engagement by team, one distribution center

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Returns	7	52
Quality	4	88
Site leadership	9	81

Try to avoid
Reporting groups below the minimum, even in a table or tooltip.

Engagement by team, one distribution center

Small teams combined into one group that clears the threshold.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Other teams (3 combined)	20	72

Recommended
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

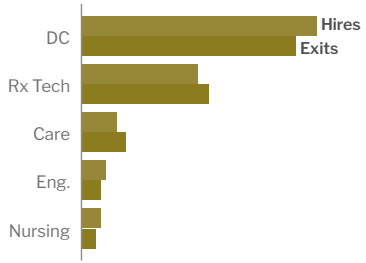
Accessible by default

About 1 in 12 men and 1 in 200 women have a color-vision deficiency. In a room of 25, assume someone does.

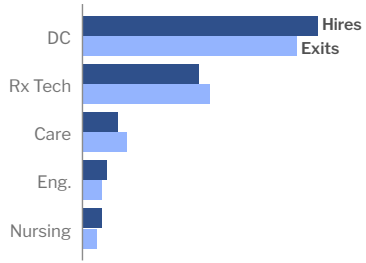
- Text contrast at least 4.5:1
- Never red vs. green; use blues and navy, with orange only for "worse"
- Never color alone: pair it with labels, position or shape
- Nothing below 9 pt on slides or 11 px on screens
- Alt text for every chart that leaves the team: the headline plus two or three key numbers

The red–green examples are simulated for deuteranopia, the most common form.

Red vs. green

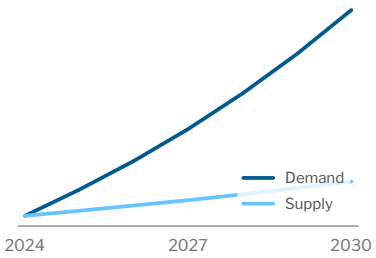


As seen with deuteranopia
Red and green hires vs. exits become one color.

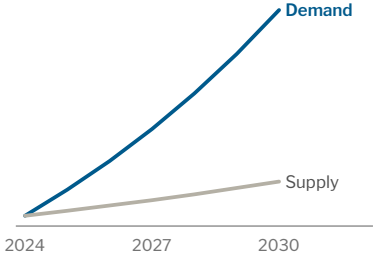


As seen with deuteranopia
Blue and light blue stay distinct.

Color as the only cue

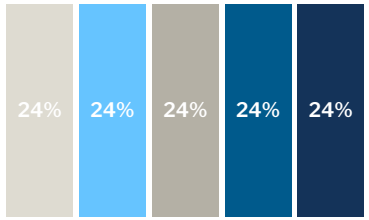


Try to avoid
The reader matches colors to a legend.

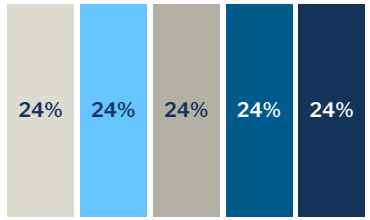


Recommended
Names at the line ends; color only reinforces.

Text on fills



Try to avoid
White on every fill fails on the light ones.



Recommended
White or navy, whichever contrasts more.

Alt text

Write it from the headline and the numbers that prove it.

> **Alt:** Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate. Pharmacy Technician 27.4%; Distribution Center Associate 24.1%; all other job families are below the enterprise rate.

In PowerPoint: right-click the chart, View Alt Text. In HTML: `aria-label` on the figure.

MAGNITUDE

Bar chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

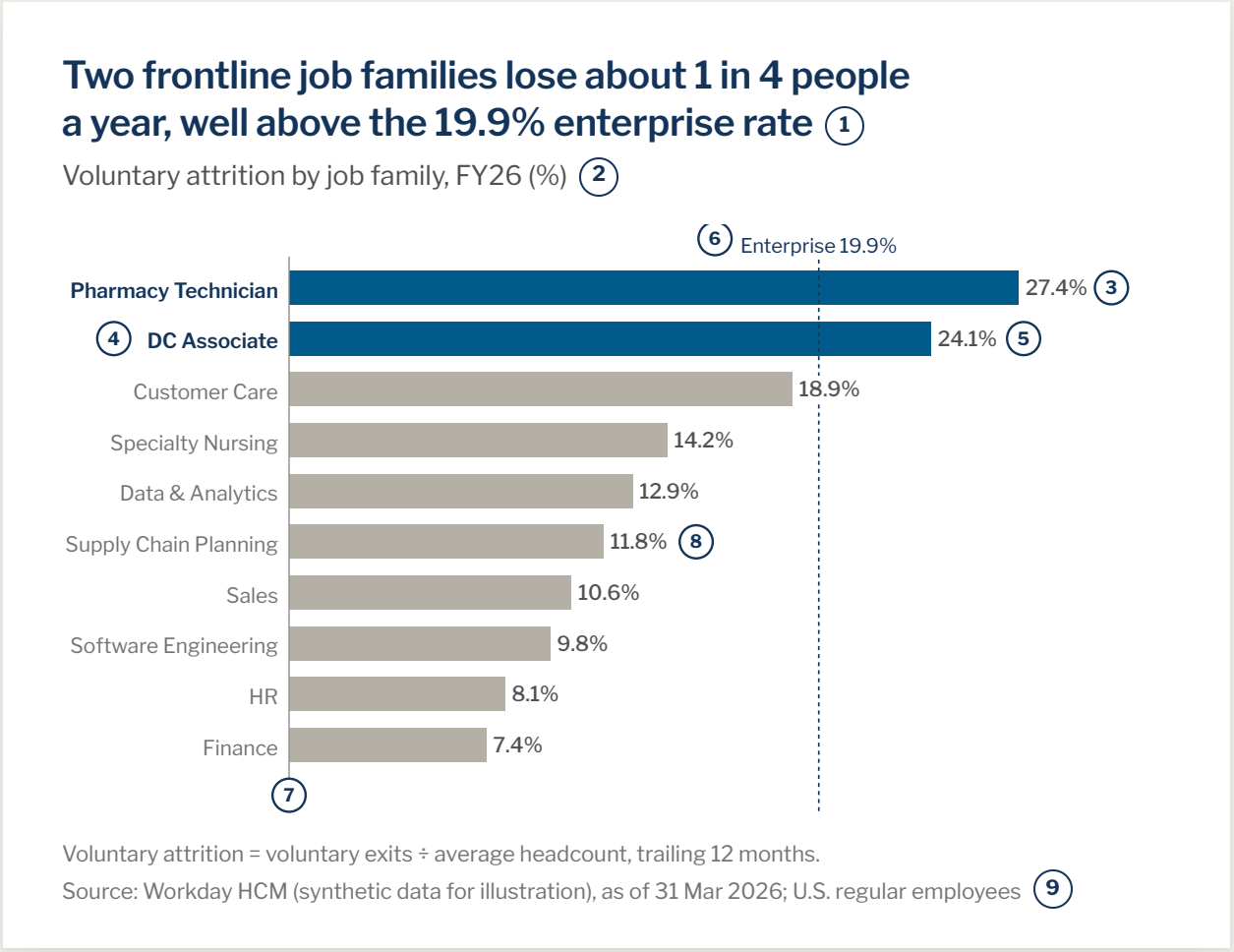
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

Bar chart variants

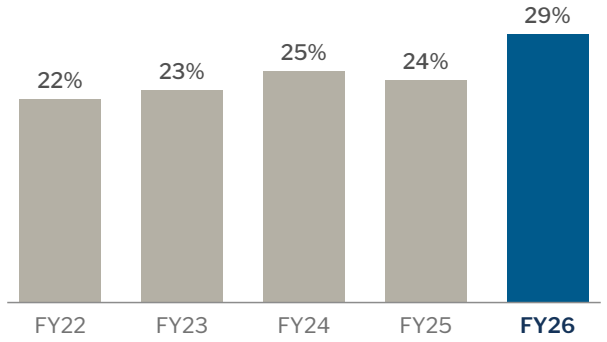
Same rules, different shapes. Pick the variant from the comparison you need to make.

When a bar is the wrong chart

- Values bunched far from zero (engagement 71–78): use a **dot plot**
- Largest more than 20× the smallest: use **proportional bubbles** or split the chart
- Parts of a whole: use a **100% stacked bar** or **waterfall**
- Many periods: use a **line**
- Spread across people: use a **distribution**

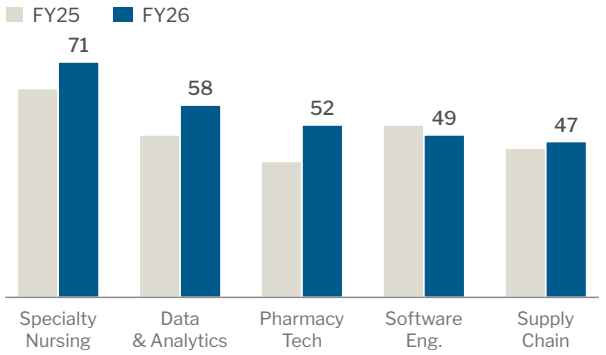
Column chart

A few periods in time order, or short category names. Here internal fill rose from 22% to 29%.



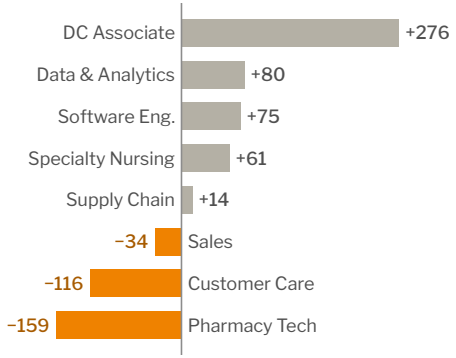
Grouped bars

Two or three series per category. Prior period in light grey, current in blue; key top-left.



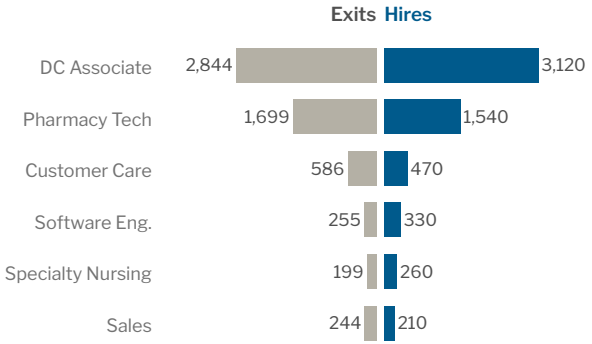
Diverging bars

Values on both sides of zero. Three job families shrank in FY26 despite hiring; negatives in orange.



Butterfly chart

Two measures back to back on the same items: FY26 exits and hires by job family.



MAGNITUDE

Bar chart best practice

Four rules that fix most bar charts.

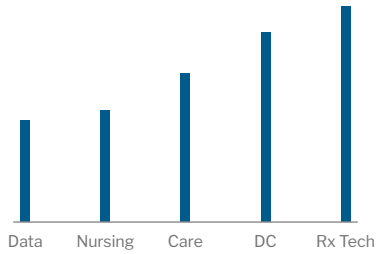
Size of bars. Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

Color of bars. A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

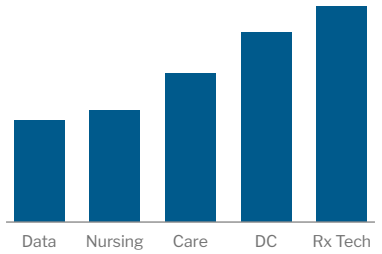
Direct labels. Gridlines and a value axis add noise. Label the bars and delete both.

Zero baseline. Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

Size of bars

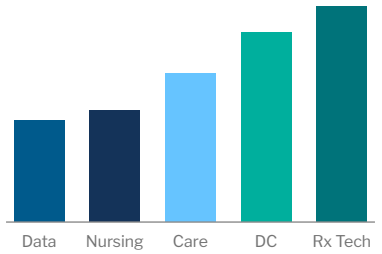


Try to avoid
Stick-thin bars; the gaps dominate.

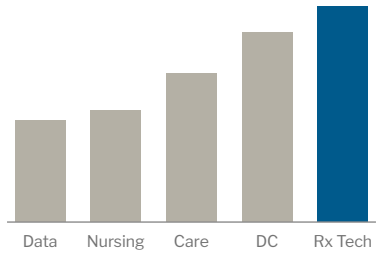


Recommended
Bar about twice the gap.

Color of bars

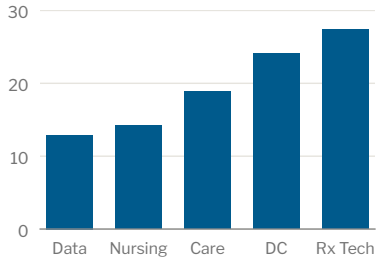


Try to avoid
A color per bar, encoding nothing.

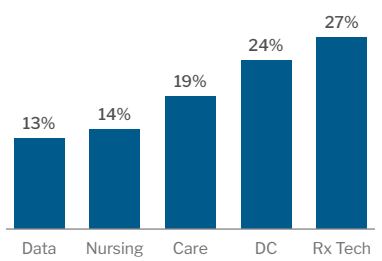


Recommended
Grey bars; blue for the one that matters.

Direct labels

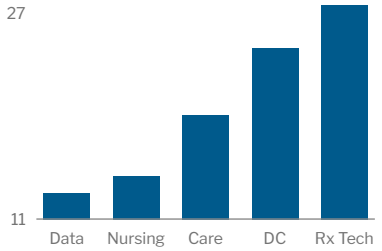


Try to avoid
Gridlines and an axis to read values from.

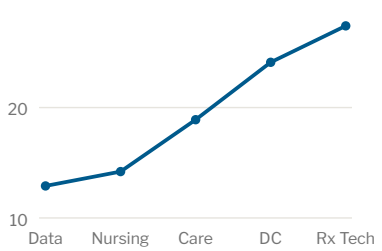


Recommended
Values on the bars; no axis, no grid.

Zero baseline



Try to avoid
A truncated axis exaggerates differences.



Recommended
A line or dot plot can use a narrow range.

Bar chart best practice, continued

Four more, from reviewing workforce decks.

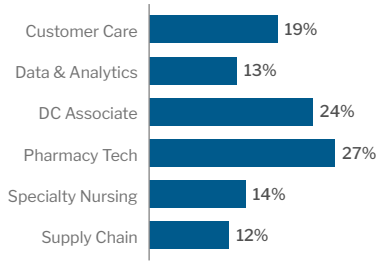
Sort with intent. Alphabetical order hides the ranking. Sort by value unless the categories have a natural order: tenure bands, job levels, survey scales.

Horizontal for long names. Rotated labels are slow to read. Job-family names need horizontal bars.

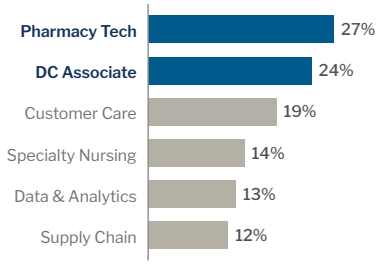
Label the benchmark in place. A reference line explained in a legend makes the reader search. Name it on the line.

Two highlights at most. Five colored bars in two colors is decoration, not emphasis.

Sort with intent

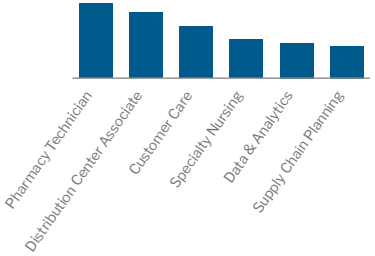


Try to avoid
Alphabetical: the ranking is hidden.

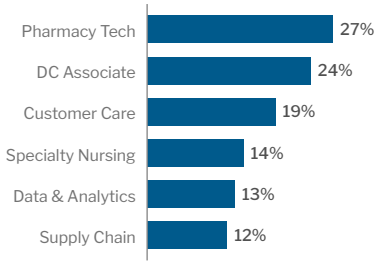


Recommended
By value, largest on top.

Horizontal for long names

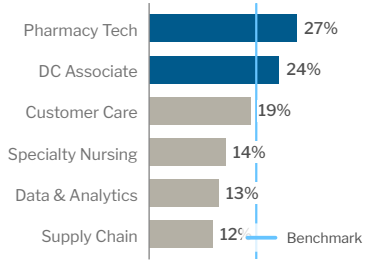


Try to avoid
Tilted names the reader has to decode.

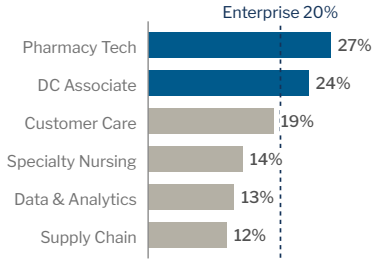


Recommended
Horizontal bars; names read naturally.

Label the benchmark

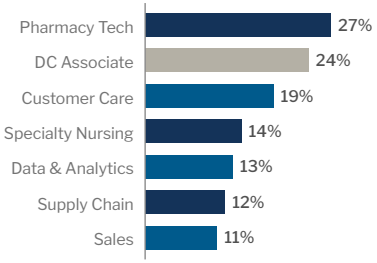


Try to avoid
A legend for one line, in a highlight color.

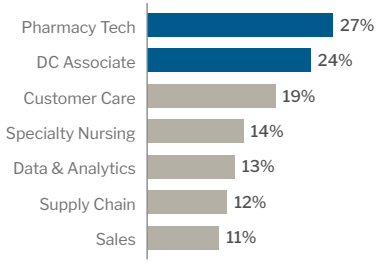


Recommended
Navy dashed line, named on the line.

Two highlights at most



Try to avoid
Everything highlighted means nothing is.



Recommended
The two the headline names.

Build a bar chart

The same chart in every tool we use. Each recipe starts from the starter kit, so the colors and fonts are already right.

Full, tested code for each platform is on the cookbook page: docs/cookbook/bar.md.

Before you send it

- [ ] Starts at zero; bar about 2× the gap
- [ ] Sorted by value or a natural order
- [ ] Values labelled; no value axis or gridlines
- [ ] At most two highlights, matching the headline
- [ ] Benchmark labelled in place
- [ ] No group under 10 people

PowerPoint and Excel	Start from the starter deck or chart workbook in `dist/office/`. Insert a clustered bar, sort the data ascending (so the largest is on top), set **Gap Width 50%**, fill all bars with Context grey and the focus bars with Focus blue. Add data labels at Outside End; delete the value axis, gridlines, legend and chart title. The slide title is the headline.
Power BI	Import `mckesson-dataviz-theme.json` and sort the bars by the measure. Drive the highlight from a measure, not by clicking bars: a `Bar color` measure returns the focus blue when attrition is above 20% and context grey otherwise, applied through **Colors › fx › Field value**. Turn on data labels; turn off the X axis and gridlines. See docs/powerbi-setup.md.
Python	matplotlib with `mckviz`; Plotly uses the same helpers (`mv.pl_highlight`).

```
bars = ax.barh(df.job_family, df.rate, height=0.67,
               color=mv.highlight(df.job_family, focus=focus))
mv.label_bars(ax, bars, fmt="{:.1f}%")
mv.clean_axes(ax, keep=("left",), grid=None); ax.set_xticks([])
mv.titles(fig, headline, subtitle); mv.source(fig, source)
mv.lint(fig)   # checks colors, zero baseline, labels vs. axis
```

HTML	Copy `templates/html/chart-card.html`: headline, subtitle and source are HTML, colors come from CSS variables such as `var(--mck-chart-focus)`, and the chart is drawn after the font loads.
R	`source("dist/r/theme_mck.R")`, then `geom_col(aes(fill = job_family), width = 0.67)` + `scale_fill_manual(values = mck_highlight(levels(df$job_family), focus = focus), guide = "none")` + `theme_mck()`.
Coding agents	"Make a McKesson bar chart of voluntary attrition by job family, FY26, highlighting the families above the enterprise rate." The `mck-chart` skill runs the workflow and returns the chart, a chart brief and alt text.